Final project report

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Github: https://github.com/madojames/STATS506/tree/main/Final%20project

Introduction to the Study and Statistical Methodology

This project investigates the gene expression profiles of two primary monocyte subsets in humans and mice to assess their similarities and differences. Using a microarray analysis approach, we try to identify differentially expressed genes between subsets in both species and evaluates conservation across species.

The data includes a matrix with probs as rows and samples as col and the number in (i,j) means the expression level of probs i in sample j, second matrix is probs and the gene name mapping relation.

Research Question

The central question addressed in this study is: *To what extent are gene expression profiles conserved between human and mouse monocyte subsets?*

We try to explore whether human-mouse models are valid analogs for studying monocyte behavior in diseases, focusing on identifying similarities and differences in gene expression to inform the use of mice as models for human immunology.

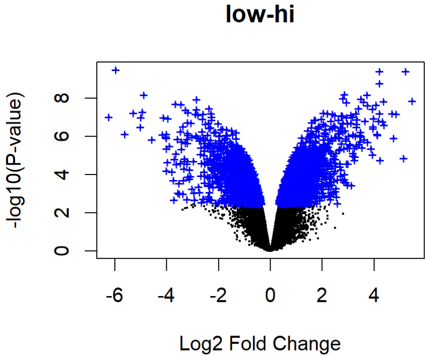
Approaches

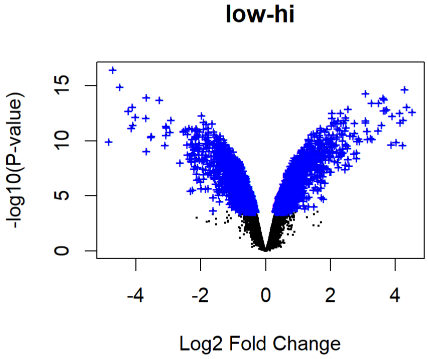
1. We cleaned and normalized the microarray data based on the expression level.
2. We compute the log fold change for each of the probs and identifies the differential expression genes by joining the two matrices.

   Approximately 270 genes in humans and 550 in mice were found to exhibit at least a two-fold difference in expression levels.
3. Using the joint list of the differential expression data to see the similarities of gene expression level in mice and human

Result

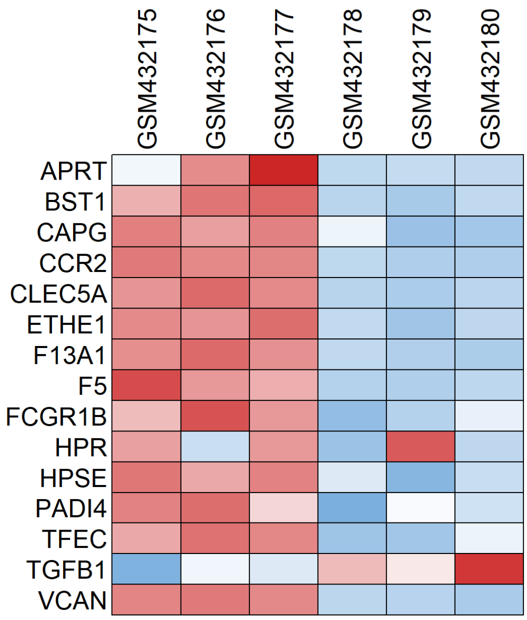
The first result is the volcano plot which reflect the overall gene expression of human (left) and mice (right).

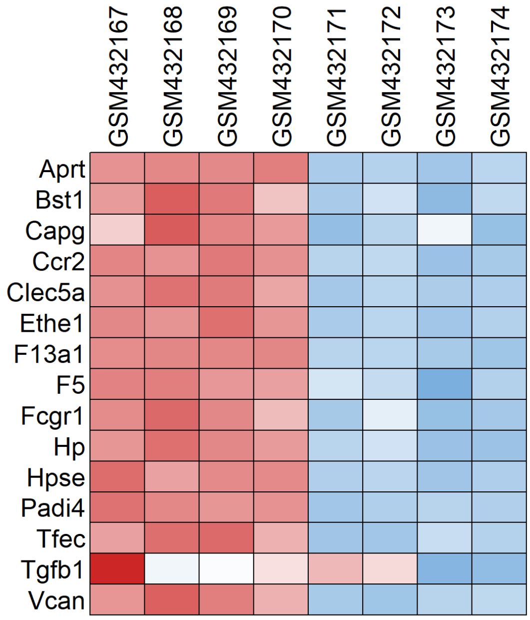




The second result is the list of difference expression genes in human and mice is ("vcan", "cd36", "f13a1", "tgfbr3", "sgms2", "tcf7l2", "plcb1", "clec5a", "ets1", "padi4", "cyfip2", "tfec", "hopx", "f5", "dusp5", "ccr2"). This might differ as gene name in mice and human is not exactly the same.

The last and the most important result is the expression pattern of the human and mice expression data. We can see the trend is similar in human and mice. These is just part of the biomarkers selected by the paper to illustrate the similarities in human and mice.





Conclusion

From the differential expression gene we got, we can see that there is difference gene group in human (271 genes) and mice (128 genes). They share some gene in common. And from part of the biological biomarkers expression level we can see that human and mice cell share a lot in common for those mentioned biomarker genes.